OMAR DESEUSA SAMPAYO

Full-Stack Developer · AI Automation

Mexico City, Mexico

PROFESSIONAL SUMMARY

Full-stack developer experienced in taking systems to production end to end: backend, frontend, infrastructure and deployment. I am the technical lead of Banco Azteca's financial education platform in Mexico, Guatemala and Honduras, where I built its administrative panels, metrics and reports from scratch. My strength is taking legacy systems and making them operable, measurable and reliable. I use AI agents (Claude Code) as a productivity multiplier, directing every technical decision and approving every production change.

PROFESSIONAL EXPERIENCE

Grupo Salinas March 2025 – Present

E-learning Technical Lead / Full-Stack Developer

Aprende y Crece · Banco Azteca's financial education platform (Mexico, Guatemala and Honduras)

- Built the administrative panel from scratch in Laravel 11 + Filament 5 (courses, modules, lessons, students, certificates, roles and permissions), letting the client manage its content without depending on development.
- Top contributor to the Mexico repository (140 of 184 commits) and sole author of the administrative panel in all three countries.
- Designed the metrics and reporting system that did not exist: a dashboard with drill-down widgets and streaming Excel export, with a single source of truth shared by panel and report.
- Built a second panel for Honduras in NestJS 11 + Angular 21 on top of a legacy database I could not modify: 8 screens, JWT authentication and SSH-tunnel access with least-privilege users.
- Fixed the password recovery flow, broken since 2020: cut response time from 95 s to 0.17 s by eliminating a linear bcrypt scan over every row.
- Delivered end to end a requirement from the Honduran banking regulator (CNBS): questionnaire audit, 3-step mobile wizard, server-side diploma locking and an email campaign to 151 graduates with zero failures.
- Recovered missing metrics: one university went from reporting 0 to 318 students; in the process I detected 46 duplicate enrollments affecting 123 people.
- Diagnosed inflated production figures caused by orphaned records (127 of 307 enrollments); fixed it without deleting data, verifying against SQL and escalating root causes to the client.
- Deployed the Guatemala installation on a server shared with 7 projects, undocumented and behind Cloudflare, with no service impact.
- Participated in implementing the development and production architecture on AWS (load balancer, WAF Captcha, S3) and in certifying the application for its go-live at IEECO.
- Produced an infrastructure sizing study based on 15 days of logs: recommended reducing from 8 vCPU / 31 GB to 2 vCPU / 4 GB and delivered technical PDF reports to the client.

Other projects

- Developed components and features in React for Portal CoE 2.0 and in Angular for other group projects; API consumption, MySQL database setup and version control with GitLab.

Proyecto Stimate September 2024 – January 2025

Full-Stack Developer

- Developed front-end features with Next.js, React and Tailwind, and REST APIs deployed on AWS.
- Designed the login and hero UX/UI in Figma; coordinated tasks with Scrum and Trello, using a Git branching workflow on GitHub.

Qodinc July 2023 – July 2024

Front-End Developer

- Developed archetypes and reusable components for several projects in Angular, Next.js and Laravel.

Krolls Telcomm de México August 2019 – September 2022

Support and NOC Engineer · Client: Nacional Monte de Piedad

- Took part in the nationwide technology refresh with Cisco Catalyst equipment; network monitoring at the NOC and ticket resolution.

PERSONAL PROJECTS

Una vez más · Inventory, sales and finance system September 2026 – Present

Product Owner and Full-Stack Developer · In production, daily use

- Migrated a real bakery business from Excel to a production web system in a few weeks: users and roles, multi-location inventory, mobile point of sale, commissions and financial reports.
- Designed the accounting model (weighted average cost, fuel cost allocation, card-terminal and salesperson commissions, partial returns) with exact decimal arithmetic, validated against the original Excel.
- End-to-end typed monorepo architecture (~17,500 lines): Angular 22, NestJS, Drizzle and PostgreSQL 18, with a shared package ensuring front and back compute identically.
- Zero-open-port infrastructure (Cloudflare Tunnel + Access), CI/CD with GitHub Actions and GHCR, SHA-based deployments, automatic migrations and encrypted daily backups.
- 300+ automated tests (Vitest; 170+ integration with Testcontainers; 88 E2E with Playwright) and WCAG 2.1 AA accessibility verified on every screen.

Retazos de Historia · Automated AI video pipeline October 2026 – Present

Creator and Developer

- Designed a pipeline that produces 1 long-form video and 3 Shorts per day for a history-focused YouTube channel, at ~2 USD per video and minutes of manual work.
- Orchestrated Claude (research and script as validated JSON), ElevenLabs (voice and music), OpenAI Images, n8n on a VPS and the YouTube Data API; programmatic rendering with Remotion (React/TypeScript) with word-synced subtitles.
- Redesigned the architecture after detecting hung n8n executions: resumable local service with atomic writes, rate limiting, retries with backoff, per-stage monitoring and idempotent publishing.

EDUCATION

Instituto Tecnológico de Tláhuac II Mexico City

Information and Communications Technologies Engineering 2013 – 2019

SKILLS, CERTIFICATIONS AND LANGUAGES

Languages: TypeScript, JavaScript, PHP 8, Python, Bash, SQL, HTML, CSS/Sass.

Frontend: Angular (signals, standalone), React, Next.js, Angular Material, Tailwind, Bootstrap, Blade, Livewire.

Backend and data: Laravel 11, Filament, NestJS, Node.js, REST, JWT, RBAC; PostgreSQL, MySQL, MariaDB, MongoDB; Drizzle, TypeORM, Eloquent.

Infrastructure and quality: Docker, AWS (EC2, ALB, WAF, S3), Cloudflare (Tunnel, Access, R2), Linux, Apache, GitHub Actions, CI/CD; Vitest, Playwright, Testcontainers, WCAG 2.1 AA.

AI and automation: Claude / Claude Code, OpenAI API, ElevenLabs, n8n, Remotion, YouTube Data API.

Methodology and tools: Scrum, Git, GitHub, GitLab, Figma, Trello.

Certifications: Full Stack Jr Web Developer JavaScript · Scrum Fundamentals Certified · AWS Skills Centers: Becoming a Cloud Practitioner.

Languages spoken: Spanish (native) · English (A2, reading technical documentation).